

## The Owl: Virtual Teleportation through XR

Alvaro Villegas\*  
Nokia

Ester Gonzalez-Sosa†  
Nokia

Pablo Perez‡  
Nokia

Juan Torres§  
Nokia

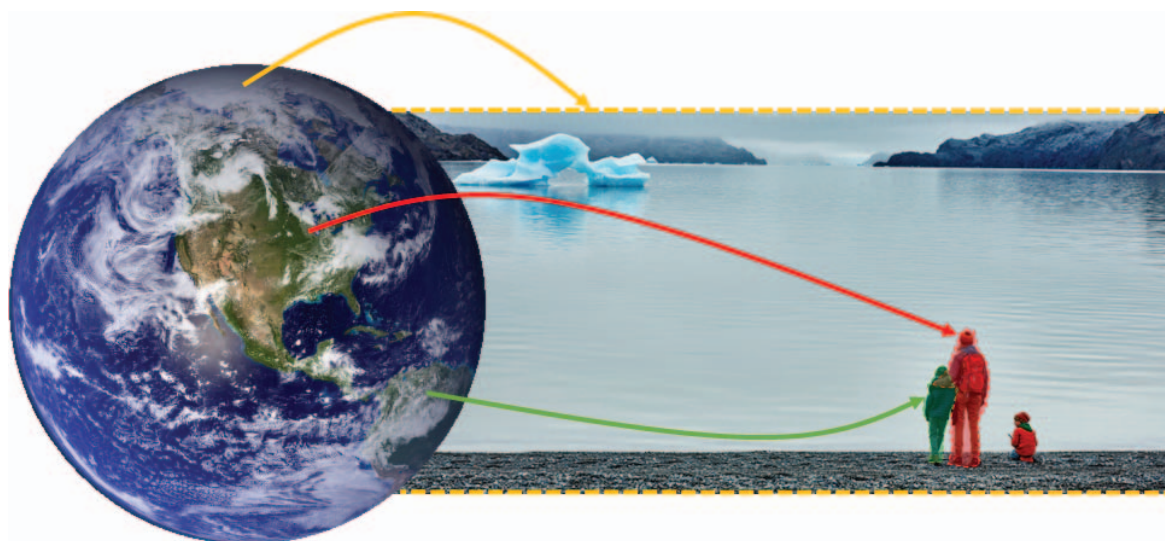


Figure 1: Virtual Teleportation Concept: people from different spots on Earth feel together in a remote, real physical place.

### ABSTRACT

We present a practical demonstration of The Owl, the prototype we have built for our concept of **Virtual Teleportation** using eXtended Reality technologies based on immersive video. We will show how a combination of state-of-the-art immersive devices (Virtual Reality goggles), a reality capture device built with off-the-shelf components and a self-designed back-end running on top of a high capable network can provide the closest experience to being teleported to a different place in the real world. Participants will not only feel present at a distant location, but they will also be able to interact with the people present there and with other remote teleported visitors. We have tried this concept in the field in many different scenarios and we are convinced it may be the foundation of the next generation human communication applications.

### 1 CONCEPT: VIRTUAL TELEPORTATION

Immersive media is the generic concept encompassing Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR) or more generically Extended Reality (XR). Historically these technologies have been used mainly for simulations, entertainment and, in the last years, also as an advanced version of social media with the recent raise of the Metaverse concept. Comparatively, VR is at a more advanced technological status than the other XR versions, and for that reason most of the mentioned applications are based on virtual environments and avatars, rather than on the real world. On a parallel track, video technology has been evolving from its initial

unidirectional nature (broadcast and on demand services) to becoming the foundation of personal communications in the last years, with videoconferencing being now universal in all society sectors and population ages. In this case the technology is purely based on real world and real people rather than simulations or avatars, but the physical limitations of regular video (essentially, a small rectangle in a screen) also limit the experience provided by its communication applications. Between both worlds, immersive video (starting from 360 video, which is slowly evolving to more volumetric versions) provides the best of both worlds: real world and real people with an immersive experience. However, so far the applications of immersive video have been limited to recordings or at most unidirectional transmission of events.

Our proposal in this area is therefore the **application of immersive video to the area of personal communications**, providing a full bidirectional capability to participants within a fully immersive experience. A reality capture device ("The Owl") is placed statically in a location and the whole space around it is captured and transmitted in real time over the network. Remote participants wearing VR goggles can then feel present at that point and interact with the people around the capture device. The resulting feeling is similar to the teleportation shown in science-fiction, in our case a "Virtual Teleportation"

Given the pervasive nature of immersivity, a communication application based on XR is by definition assymmetric: the place where the reality capture device is placed becomes the main site, to where remote participants will feel teleported. The physical surroundings of remote participants is less relevant, although in many cases it may be interesting to keep part of it, including their own bodies. This also presents some challenges for the "reverse presence": the local people around the capture device should have some feeling of presence of the remote participants with whom they are interacting. We have experimented with different ways of implementing this path; one of them is based on avatars on a screen, which will be the one shown in the demonstration.

\*e-mail: alvaro.villegas@nokia.com

†e-mail: ester.gonzalez@nokia.com

‡e-mail: pablo.perez@nokia.com

§e-mail: juan.torresarjona@nokia.com

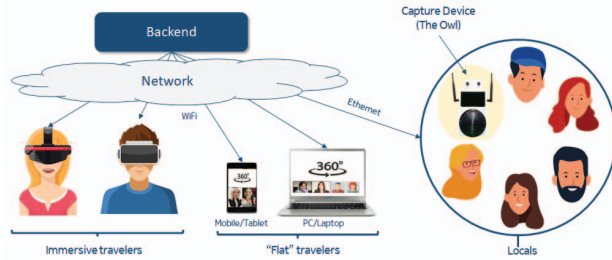


Figure 2: Architecture of the Virtual Teleportation demonstration

## 2 ARCHITECTURE AND EXPERIENCE

Our technology enables a communication session with peers all around the world, but for this demonstration our idea is to have everybody in the same venue (with some distance to avoid direct contact). In particular, there will be two groups of users:

- **Locals:** people around the capture device (called "the locals") simply experience a regular real life meeting, with the only difference that some people that participate in the session are not physically there. With the initial reverse presence technology available now, this will be similar to having a screen with some people present in videoconferencing.
- **Travelers:** this group consists of an arbitrary number of remote participants (1 to 5, depending on the availability of VR goggles), called the "travelers". Each of them will be physically located in an isolated place in the venue, although the prototype enables joining from any place in the world.

The most relevant experience to be shown is for the travelers. A traveler sits down and puts on the VR goggles (Meta Quest2 or Quest3). When the application is launched each traveler feels immediately present in the exact spot where the Owl is located. They can look around in all directions (3 degrees of freedom), hear everything, and also talk freely with the people around the Owl. In this initial version the local scenario around the travelers disappears (as in any VR application) but a virtual representation of their hands in real time provides a useful hook to their own body.

The locals can see a representation of travelers through cartoonish avatar heads (customizable with the VR application) on a screen attached to the Owl. The heads move in real time according to the real movement of the travelers they represent. Avatar lips move according to the travelers voices, and their hands are also represented through avatars in this screen.

There may be multiple travelers. When this happens, each traveler sees other travelers as avatar heads floating in the scene. The communication between travelers is also possible, everybody in the session can communicate freely with all the rest. We are currently working on more advanced versions in which the avatars are placed in appropriate empty places in the scene (detected through AI) instead of floating around.

While the best experience is achieved using VR goggles, it is also possible for a traveler to join the communication session using a regular "flat" device (mobile phone, tablet). In this case the experience is not immersive, but the scene can still be accessed rotating manually the view in the touchscreen. The selfie camera in the device is used to capture the user as in videoconferences; this video is then added as a floating rectangle instead of the avatar for this user.

## 3 REQUIREMENTS

What we need for setting up and running the demonstration of our Virtual Teleportation is the following:

- A clear space of around 4 to 8 square meters to place the capture device (the Owl), on a tripod that we will provide. It should have good lighting, and at least an AC socket. Ambient noise should be as low as possible. Furniture or any physical elements present in the space will be welcome. In this place we need an Ethernet connection to a network with at least 15Mbps of sustained throughput.
- At least two separate places, out of audio reach from each other and from the capture place. In each of these places we need AC power and a good WiFi access (at least 15Mbps, sustained) to the same network.

We will provide the capture device, all required wiring and the VR goggles.

Optionally, a richer demonstration may be offered if these elements are available:

- Internet access with good throughput (at least 30Mbps bidirectional) in the network that connects all spots. This access should not have any firewall preventing a VPN to work.
- Some users at other locations onsite or at other remote places willing to join the experience. They will need Quest2 or Quest3 VR goggles with an application that we will provide. Some prior setup will be required for them.

## 4 APPLICATIONS

We consider the demonstrated technology as the seed for a whole new set of communication services that may be applied to multiple sectors:

- **Education:** educational applications with a strong practical component, in which being physically present in the classroom is relevant the Virtual Teleportation may enable remote students to get the same education quality as the locals. In some cases it may also be possible to use it for having the teacher remote.
- **Remote Expert:** the remote assistance of an expert in a difficult situation may be much more efficient if the expert feels fully present at the location where the assistance is required. This includes technical assistance, first responders, surgery, etc.
- **Healthcare:** children or in general patients at hospitals or Elderly in nursing homes may have some relief from their confinement by virtually teleporting to an outdoors scenario or simply to their relatives home.
- **Tourism:** visiting a remote place and interacting with people in real time there may provide a very powerful touristic experience.
- **Meetings:** regular work meetings may be enriched with the immersive nature of this application, what eventually sets remote participants in the same condition as the locals present in the meeting room (no remote isolation feeling).

## 5 CONCLUSIONS

With this demonstration we want to show how XR technology can be applied to a whole new world of applications in the real world, human-centric and focused on communications.

## ACKNOWLEDGMENTS

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